

Introduction to Unsupervised Learning

What is Unsupervised Learning?

Definition:

- Learning from data without labels
- Dataset: $\mathcal{D} = \{x_1, x_2, \dots, x_n\}$, $x_i \in \mathbb{R}^d$

Goal:

- Discover hidden structure
- Identify patterns and relationships

Core Tasks

Clustering

- K-Means
- Hierarchical Clustering
- DBSCAN

Dimensionality Reduction

- PCA
- t-SNE
- Autoencoders

Density Estimation

- Gaussian Mixture Models
- Kernel Density Estimation

Supervised vs Unsupervised

Aspect	Supervised	Unsupervised
Labels	Present	Absent
Objective	Predict y	Discover structure
Evaluation	Accuracy, F1	Inertia, Silhouette

K-Means Clustering

Goal:

- Partition data into K clusters

Objective Function:

$$J = \sum_{k=1}^K \sum_{x_i \in C_k} ||x_i - \mu_k||^2$$

K-Means Algorithm

- 1 Initialize centroids randomly
- 2 Assign points to nearest centroid
- 3 Update centroids
- 4 Repeat until convergence

Assignment Step:

$$\text{cluster}(x_i) = \arg \min_k ||x_i - \mu_k||$$

Update Step:

$$\mu_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$$

Worked Example: Dataset

Points:

- $x_1 = (1, 1)$
- $x_2 = (1.5, 2)$
- $x_3 = (3, 4)$
- $x_4 = (5, 7)$

Let: $K = 2$

Step 1: Initialization

Initial Centroids:

- $\mu_1 = (1, 1)$
- $\mu_2 = (5, 7)$

Step 2: Assignment

Distance Example:

For $x_2 = (1.5, 2)$:

$$d(x_2, \mu_1) = \sqrt{(1.5 - 1)^2 + (2 - 1)^2}$$

$$d(x_2, \mu_2) = \sqrt{(1.5 - 5)^2 + (2 - 7)^2}$$

Clusters:

- $C_1 = \{(1, 1), (1.5, 2)\}$
- $C_2 = \{(3, 4), (5, 7)\}$

Step 3: Update Centroids

$$\mu_1 = \left(\frac{1 + 1.5}{2}, \frac{1 + 2}{2} \right) = (1.25, 1.5)$$

$$\mu_2 = \left(\frac{3 + 5}{2}, \frac{4 + 7}{2} \right) = (4, 5.5)$$

Convergence

- Repeat assignment and update steps
- Stop when clusters do not change

Assumptions:

- Spherical clusters
- Equal variance

Limitations:

- Need to choose K
- Sensitive to initialization
- Affected by outliers

Choosing K

Elbow Method:

- Plot K vs objective J
- Look for a bend (elbow point)

Summary

- Unsupervised learning finds hidden structure
- K-Means minimizes intra-cluster variance
- Widely used for clustering tasks